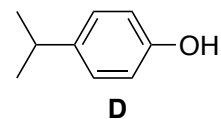
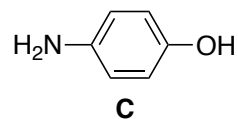
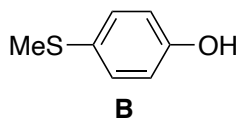
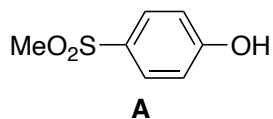
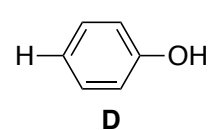
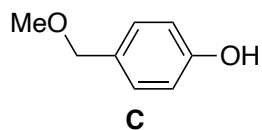
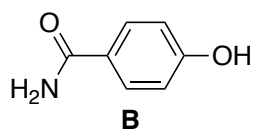
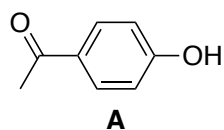


Ranking Acidity of Substituted Phenols

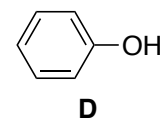
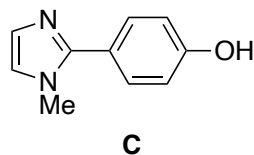
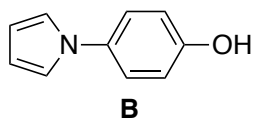
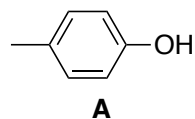
1. Rank the phenols below from most acidic (1) to least acidic (4).



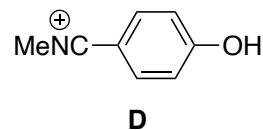
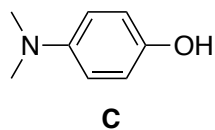
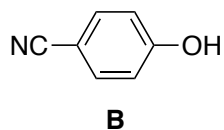
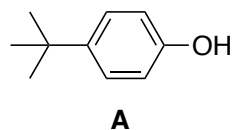
2. Rank the phenols below from most acidic (1) to least acidic (4).



3. Rank the phenols below from most acidic (1) to least acidic (4).



4. Rank the phenols below from most acidic (1) to least acidic (4).



Ranking Acidity of Substituted Phenols | KEY

1. A, D, B, C. Draw a Lewis structure for the methanesulfonyl ($\text{MeO}_2\text{S}-$) group to convince yourself that it's electron withdrawing.
2. A, B, D, C *or* A, B, C, D. The methoxymethyl group is actually electron withdrawing overall, due to the inductive effect of the alkoxy oxygen. I wouldn't expect you to predict this but would be happy if you did!
3. C, D, A, B. Note that the *N*-pyrrolyl substituent in **B** is electron donating, but the 2-imidazolyl substituent in **C** is withdrawing.
4. D, B, A, C. Methylation of the cyano group increases its electron-withdrawing power *massively*. Your chemical intuition should tell you that, *ceteris paribus*, cations are more acidic than neutral molecules.